

Task 1

Task1

```
LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ envsubst '${NS}' < manifests/guardrails.yaml | kubectl -n "${NS}" apply -f -
resourcequota/team-budget unchanged
networkpolicy/networking.k8s.io/private-object-store unchanged
serviceaccount/observer unchanged
role.rbac.authorization.k8s.io/observer unchanged
rolebinding.rbac.authorization.k8s.io/observer unchanged

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "${NS}" get resourcequota -o yaml > evidence/quota-before.yaml

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ envsubst '${NS}' ${CLIENT_IMAGE}' < manifests/positive.yaml | kubectl -n "${NS}" create -f -
pod/quota-positive created

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "${NS}" get pod quota-positive
NAME          READY   STATUS    RESTARTS   AGE
quota-positive 0/1     Completed 2 (29s ago) 41s

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "${NS}" delete pod quota-positive --wait=true
pod "quota-positive" deleted

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ envsubst '${NS}' ${CLIENT_IMAGE}' < manifests/negative.yaml | kubectl -n "${NS}" create -f - > evidence/quota-reject.txt 2>&1

LENOVO@DESKTOP-NK08GJ2 MINGW64 ~/bigdata-lab01-group04 (main)
$ cat evidence/quota-reject.txt
Error from server (Forbidden): error when creating "STDIN": pods "quota-negative" is forbidden: exceeded quota: team-budget, requested: requests.cpu=3, used: requests.cpu=0, limited: requests.cpu=2
```

Task 2

Task2

```
MINGW64/c/Users/LENOVO/bigdata-lab01-group04
Total 14 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 1 local object.
To https://github.com/Hoang1711provip/bigdata-lab01-group04.git
   5cf5f63..611318f  main    -> main

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ export NS="group04"

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "$NS" create secret generic s3-config \
  --from-literal=s3.json="{
    \"identities\": [
      {
        \"name\": \"owner\",
        \"credentials\": [{\"accessKey\": \"ownerAK\", \"secretKey\": \"ownerSK123\"}],
        \"actions\": [\"Admin\", \"Read\", \"Write\"]
      },
      {
        \"name\": \"ingestor\",
        \"credentials\": [{\"accessKey\": \"ingestorAK\", \"secretKey\": \"ingestorSK123\"}],
        \"actions\": [\"Read\", \"Write\"]
      },
      {
        \"name\": \"analyst\",
        \"credentials\": [{\"accessKey\": \"analystAK\", \"secretKey\": \"analystSK123\"}],
        \"actions\": [\"Read\"]
      }
    ]
  }"
--dry-run=client -o yaml | kubectl apply -f -
secret/s3-config created

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "$NS" create secret generic owner-creds \
  --from-literal=accessKey=ownerAK \
  --from-literal=secretKey=ownerSK123 \
  --dry-run=client -o yaml | kubectl apply -f -
secret/owner-creds created

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "$NS" create secret generic ingestor-creds \
  --from-literal=accessKey=ingestorAK \
  --from-literal=secretKey=ingestorSK123 \
  --dry-run=client -o yaml | kubectl apply -f -
secret/ingestor-creds created

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "$NS" create secret generic analyst-creds \
  --from-literal=accessKey=analystAK \
  --from-literal=secretKey=analystSK123 \
  --dry-run=client -o yaml | kubectl apply -f -
secret/analyst-creds created

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ git pull origin main

MINGW64/c/Users/LENOVO/bigdata-lab01-group04
$ git pull origin main
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.63 KiB | 111.00 KiB/s, done.
From https://github.com/Hoang1711provip/bigdata-lab01-group04
   * branch      main      -> FETCH_HEAD
   611318f..bccdf33  main    -> origin/main
Updating 611318f..bccdf33
Fast-forward
  manifests/store.yaml | 97 ++++++-----
  1 file changed, 33 insertions(+), 64 deletions(-)

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ export STORAGE_IMAGE="chrislusf/seaweedfs:latest"

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ envsubst '$(NS) ${STORAGE_IMAGE}' < manifests/store.yaml | kubectl -n "$NS" apply -f -
kubectl -n "$NS" rollout status deployment/objects
persistentvolumeclaim/object-data created
deployment.apps/objects created
service/objects created
waiting for deployment "objects" rollout to finish: 0 of 1 updated replicas are available...
deployment "objects" successfully rolled out

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ cat <<EOF | kubectl -n "$NS" apply -f -
ap -
apiVersion: v1
kind: Pod
metadata:
  name: client-owner
  command: ["sleep", "3600"]
bash: -: command not found
bash: apiVersion: command not found
bash: kind: command not found
bash: metadata: command not found
bash: name: command not found
bash: command: command not found

LENOVO@DESKTOP-MK08632 MINGW64 ~/bigdata-lab01-group04 (main)
$ cat <<EOF | kubectl -n "$NS" apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: client-owner
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
```

MINOW64/c/Users/LENOVO/bigdata-lab01-group04

```
bash: -: command not found
bash: apiVersion:: command not found
bash: kind:: command not found
bash: metadata:: command not found
bash: name:: command not found
bash: command:: command not found
```

```
LENOVO@DESKTOP-MK086J2 MINOW64 ~/bigdata-lab01-group04 (main)
```

```
$ cat <<EOF | kubectl -n "$NS" apply -f -
```

```
apiVersion: v1
kind: Pod
metadata:
  name: client-owner
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    envFrom:
    - secretRef:
        name: owner-creds
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: client-ingestor
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    envFrom:
    - secretRef:
        name: ingestor-creds
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: client-analyst
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    envFrom:
    - secretRef:
        name: analyst-creds
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: client-analyst
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    envFrom:
    - secretRef:
        name: analyst-creds
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: client-blocked
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    envFrom:
    - secretRef:
        name: analyst-creds
```

```
LENOVO@DESKTOP-MK086J2 MINOW64 ~/bigdata-lab01-group04 (main)
```

```
$ cat <<EOF | kubectl -n "$NS" apply -f -
```

```
apiVersion: v1
kind: Pod
metadata:
  name: client-owner
  namespace: $NS
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    resources:
      requests:
        cpu: 50m
        memory: 64Mi
      limits:
        cpu: 100m
        memory: 128Mi
    envFrom:
```

```
LinuxVagrant@vagrant:~/bigdata-lab01-group04 (main)
$ cat <EOF | kubectl -n "${NS}" apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: client-owner
  namespace: ${NS}
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    resources:
      requests:
        cpu: 50m
        memory: 64Mi
      limits:
        cpu: 100m
        memory: 128Mi
    envFrom:
    - secretRef:
        name: owner-creds
---
apiVersion: v1
kind: Pod
metadata:
  name: client-ingestor
  namespace: ${NS}
spec:
  automountServiceAccountToken: false
  containers:
  - name: client
    image: alpine
    command: ["sleep", "3600"]
    resources:
      requests:
        cpu: 50m
        memory: 64Mi
      limits:
        cpu: 100m
        memory: 128Mi
    envFrom:
    - secretRef:
        name: ingestor-creds
---
apiVersion: v1
kind: Pod
metadata:
  name: client-analyst
  namespace: ${NS}
spec:
  automountServiceAccountToken: false
  containers:
```

```

MINGW64/C:/Users/LENOVO/Desktop/lab01-group04
spec:
  automountServiceAccountToken: false
  containers:
    - name: client
      EOP memory: 128Mi, "3600"]: false
pod/client-owner created
pod/client-ingestor created
pod/client-analyst created
pod/client-blocked created
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP              NODE               NOMINATED NODE   READINESS GATES
client-analyst 1/1     Running   0           39s   10.244.0.10     desktop-control-plane <none>             <none>
client-blocked 1/1     Running   0           39s   10.244.0.11     desktop-control-plane <none>             <none>
client-ingestor 1/1     Running   0           39s   10.244.0.9      desktop-control-plane <none>             <none>
client-owner   1/1     Running   0           39s   10.244.0.8      desktop-control-plane <none>             <none>
objects-7db59fd589-q2tfm 1/1     Running   0           39s   10.244.0.7      desktop-control-plane <none>             <none>
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" get pods,pvc,svc -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP              NODE               NOMINATED NODE   READINESS GATES
pod/client-analyst 1/1     Running   0           116s   10.244.0.10     desktop-control-plane <none>             <none>
pod/client-blocked 1/1     Running   0           116s   10.244.0.11     desktop-control-plane <none>             <none>
pod/client-ingestor 1/1     Running   0           116s   10.244.0.9      desktop-control-plane <none>             <none>
pod/client-owner   1/1     Running   0           116s   10.244.0.8      desktop-control-plane <none>             <none>
pod/objects-7db59fd589-q2tfm 1/1     Running   0           6m48s  10.244.0.7      desktop-control-plane <none>             <none>
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE   SELECTOR
service/objects ClusterIP      10.96.11.138    <none>           8333/TCP         6m48s app=objects
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" get pod,pvc,svc,secret -o wide > evidence/store-status.txt
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" describe pvc object-data > evidence/pvc-detail.txt
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" get pod -l app=objects -o jsonpath='{.Pod UID: "}{.items[0].metadata.uid}"\nImageID: "}{.items[0].status.containerStatuses[0].imageID}"\n"} > evidence/store-identifiers.txt
kubectl -n "SNS" get pvc object-data -o jsonpath='{.PVC UID: "}{.metadata.uid}"\nstorageClass: "}{.spec.storageClassName}"\n"} > evidence/store-identifiers.txt
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ kubectl -n "SNS" Top -l app=objects > evidence/store-logs.txt 2>&1
LENOVODESKTOP-ND8G32 MINGW64 ~/bigdata-lab01-group04 (main)
$ git add evidence/
warning: in the working copy of 'evidence/resource-samples.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'evidence/pvc-detail.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'evidence/store-identifiers.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'evidence/store-logs.txt', LF will be replaced by CRLF the next time Git touches it

```

Task 3

Task3

```
DELL@DESKTOP-UHCRB1I MINGW64 ~/lab
$ rm -f evidence/security/s3-results.csv
t() {
  id=$1; pod=$2; exp=$3; shift 3
  f=evidence/security/$id.json
  kubectl -n "$NS" exec "$pod" -- python /opt/s3lab.py probe "$@" > "$f" 2>/dev/null
  http=$(grep -o "http": [A.]* "$f" | cut -d' ' -f2)
  err=$(grep -o "error": [A.]* "$f" | cut -d' ' -f2 | tr -d '"')
  if [ "$exp" = Allow ]; then
    [ "$http" = 200 ] && res=pass || res=FAIL
  elif [ "$http" = 403 ] && [ "$err" = AccessDenied ]; then res=pass
  elif [ "$http" = 200 ]; then res=FAIL-forbidden-op-succeeded
  else res=inconclusive; fi
  echo "$id,$pod,$exp,$http,$err,$res,$f" | tee -a evidence/security/s3-results.csv
}

t S01 ingestor Allow put research-raw auth-probe.txt
t S02 ingestor Allow get research-raw fixture.txt
t S03 ingestor Allow list research-raw ""
t S04 ingestor Deny put research-release auth-probe.txt
t S05 ingestor Deny get research-release fixture.txt
t S06 analyst Allow get research-release fixture.txt
t S07 analyst Allow list research-release ""
t S08 analyst Deny put research-release auth-probe.txt
t S09 analyst Deny delete research-release delete-probe.txt
t S10 analyst Deny get research-raw fixture.txt
t S11 analyst Deny list research-raw ""
t S12 owner Deny get research-release fixture.txt anon
S01,ingestor,put research-raw auth-probe.txt,Allow,http=200/null,pass,evidence/security/S01.json
S02,ingestor,get research-raw fixture.txt,Allow,http=200/null,pass,evidence/security/S02.json
S03,ingestor,list research-raw,Allow,http=200/null,pass,evidence/security/S03.json
S04,ingestor,put research-release auth-probe.txt,Deny,http=403/AccessDenied,pass,evidence/security/S04.js
on
S05,ingestor,get research-release fixture.txt,Deny,http=403/AccessDenied,pass,evidence/security/S05.json
S06,analyst,get research-release fixture.txt,Allow,http=200/null,pass,evidence/security/S06.json
S07,analyst,list research-release,Allow,http=200/null,pass,evidence/security/S07.json
S08,analyst,put research-release auth-probe.txt,Deny,http=403/AccessDenied,pass,evidence/security/S08.js
on
S09,analyst,delete research-release delete-probe.txt,Deny,http=403/AccessDenied,pass,evidence/security/S
09.json
S10,analyst,get research-raw fixture.txt,Deny,http=403/AccessDenied,pass,evidence/security/S10.json
S11,analyst,list research-raw,Deny,http=403/AccessDenied,pass,evidence/security/S11.json
S12,owner,get research-release fixture.txt anon,Deny,http=403/AccessDenied,pass,evidence/security/S12.js
on
```

```
DELL@DESKTOP-UHCRB1I MINGW64 ~/lab
$ python observer.py
OBS=private/observer.json
POD=$(kubectl -n "$NS" get pod -l app=objects -o jsonpath='{.items[0].metadata.name}')
kubectl -n "$NS" run quota-positive --image=s3lab:1 --restart=Never --dry-run=client -o yaml > positive.
yaml

r() { id=$1; shift; "$@" > evidence/security/$id.out 2> evidence/security/$id.err; echo "$id exit=$?"; }

r K01 kubectl --kubeconfig="$OBS" get pods
r K02 kubectl --kubeconfig="$OBS" get events
r K03 kubectl --kubeconfig="$OBS" logs "$POD"
r K04 kubectl --kubeconfig="$OBS" get secret s3-config
r K05 kubectl --kubeconfig="$OBS" create -f positive.yaml --dry-run=server
r K06 kubectl --kubeconfig="$OBS" delete pod "$POD" --dry-run=server
head -c 300 evidence/security/K04.err; echo
head -c 300 evidence/security/K05.err; echo
head -c 300 evidence/security/K06.err; echo
K01 exit=0
K02 exit=0
K03 exit=0
K04 exit=1
K05 exit=1
K06 exit=1
Error from server (Forbidden): secrets "s3-config" is forbidden: User "system:serviceaccount:group04:obs
erver" cannot get resource "secrets" in API group "" in the namespace "group04"

Error from server (Forbidden): error when creating "positive.yaml": pods is forbidden: User "system:serv
iceaccount:group04:observer" cannot create resource "pods" in API group "" in the namespace "group04"

Error from server (Forbidden): pods "objects-6db89bb7c7-2btxd" is forbidden: User "system:serviceaccount
:group04:observer" cannot delete resource "pods" in API group "" in the namespace "group04"
```

```

DELL@DESKTOP-UHCRB1I MINGW64 ~/lab
$ r() { kubectl -n "$NS" exec "$1" -- python /opt/s3lab.py tcp "$2" "$3" | tee evidence/security/$4.json ; }
r owner objects 8333 N01-before
r blocked objects 8333 N02
POD_IP=$(kubectl -n "$NS" get pod -l app=objects -o jsonpath='{.items[0].status.podIP}')
r owner "$POD_IP" 8888 N03
r owner objects 8333 N01-after
kubectl -n "$NS" logs deploy/objects | grep -i 8888 > evidence/security/N03-listening.txt
wc -c evidence/security/N03-listening.txt
{"host": "objects", "port": 8333, "tcp_connected": true, "utc": "2026-10-04T08:14:08.330411+00:00"}
{"error": "TimeoutError", "host": "objects", "port": 8333, "seconds": 3.00610599599986, "tcp_connected": false, "utc": "2026-10-04T08:14:12.336502+00:00"}
{"error": "TimeoutError", "host": "10.244.0.6", "port": 8888, "seconds": 3.0028403119999894, "tcp_connected": false, "utc": "2026-10-04T08:14:16.795599+00:00"}
{"host": "objects", "port": 8333, "tcp_connected": true, "utc": "2026-10-04T08:14:17.716375+00:00"}
44 evidence/security/N03-listening.txt

```

```

DELL@DESKTOP-UHCRB1I MINGW64 ~/lab
$ kubectl create namespace outsider
kubectl -n outsider run probe --image=s3lab:1 --image-pull-policy=IfNotPresent --restart=Never --command -- sleep infinity
kubectl -n outsider wait --for=condition=Ready pod/probe --timeout=60s
kubectl -n outsider exec probe -- python -c "import socket; print(socket.gethostbyname('objects.$NS.svc.cluster.local'))" | tee evidence/security/N04-dns.txt
kubectl -n outsider exec probe -- python /opt/s3lab.py tcp kube-dns.kube-system.svc.cluster.local 53 | tee evidence/security/N04-routing.json
kubectl -n outsider exec probe -- python /opt/s3lab.py tcp objects.$NS.svc.cluster.local 8333 | tee evidence/security/N04.json
namespace/outside created
pod/probe created
pod/probe condition met
10.96.129.146
{"host": "kube-dns.kube-system.svc.cluster.local", "port": 53, "tcp_connected": true, "utc": "2026-10-04T08:16:09.414324+00:00"}
{"error": "TimeoutError", "host": "objects.group04.svc.cluster.local", "port": 8333, "seconds": 3.006189035999796, "tcp_connected": false, "utc": "2026-10-04T08:16:13.182813+00:00"}

```

Task 5

Task5

```
MINGW64:/c/Users/DELL/BigData_Lab01_Group4

DELL@DESKTOP-UHCRB11 MINGW64 ~
$ cd /c/Users/DELL/BigData_Lab01_Group4

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py watch 300 > evidence/canary.jsonl
Error from server (NotFound): pods "ingestor" not found

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ export NS=group04 MSYS_NO_PATHCONV=1

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ $ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py watch 300 > evidence/canary.jsonl
bash: $: command not found

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py watch 300 > evidence/canary.jsonl

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$
```

```
DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py probe list research-raw bench/r1-cl | tee /c/Users/DELL/Lab/evidence/bench-listing.json
{"bucket": "research-raw", "bytes": 0, "errors": null, "http": 200, "key": "bench/r1-cl", "keys": [{"bench/r1-cl/000.bin", "bench/r1-cl/001.bin", "bench/r1-cl/002.bin", "bench/r1-cl/003.bin", "bench/r1-cl/004.bin", "bench/r1-cl/005.bin", "bench/r1-cl/006.bin", "bench/r1-cl/007.bin", "bench/r1-cl/008.bin", "bench/r1-cl/009.bin", "bench/r1-cl/010.bin", "bench/r1-cl/011.bin", "bench/r1-cl/012.bin", "bench/r1-cl/013.bin", "bench/r1-cl/014.bin", "bench/r1-cl/015.bin", "bench/r1-cl/016.bin", "bench/r1-cl/017.bin", "bench/r1-cl/018.bin", "bench/r1-cl/019.bin", "bench/r1-cl/020.bin", "bench/r1-cl/021.bin", "bench/r1-cl/022.bin", "bench/r1-cl/023.bin", "bench/r1-cl/024.bin", "bench/r1-cl/025.bin", "bench/r1-cl/026.bin", "bench/r1-cl/027.bin", "bench/r1-cl/028.bin", "bench/r1-cl/029.bin", "bench/r1-cl/030.bin", "bench/r1-cl/031.bin"}, {"ms": 22.767804000997776, "ok": true, "op": "list", "principal": "ingestor", "request_id": "18084868830c22008674457c", "utc": "2026-10-04T10:03:09.873272+00:00"}]

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" get pod -l app=objects -o yaml > evidence/pod-before.yaml
$ kubectl -n "SNS" get pvc object-data -o yaml > evidence/pvc-before.yaml

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ POD=$(kubectl -n "SNS" get pod -l app=objects -o jsonpath='{.items[0].metadata.name}')
$ date -u +%FT%T%Z > evidence/delete-time.txt
$ kubectl -n "SNS" delete pod "$POD" --wait=false
$ kubectl -n "SNS" wait --for=delete pod/"$POD" --timeout=120s
$ kubectl -n "SNS" rollout status deployment/objects --timeout=120s
$ AC

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$ kubectl -n "SNS" wait --for=delete pod/"$POD" --timeout=120s
$ kubectl -n "SNS" rollout status deployment/objects --timeout=120s
pod "objects-6db89bb7c7-2btxd" deleted from group04 namespace
pod/objects-6db89bb7c7-2btxd condition met
waiting for deployment "objects" rollout to finish: 0 of 1 updated replicas are available...
deployment "objects" successfully rolled out

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" get pod -l app=objects -o yaml > evidence/pod-after.yaml
$ kubectl -n "SNS" get pvc object-data -o yaml > evidence/pvc-after.yaml

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py verify bench/r1-cl > evidence/after-recovery.jsonl

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" get events --sort-by='.metadata.creationTimestamp' > evidence/events.log
$ kubectl -n "SNS" logs -l app=objects --all-containers=true --tail=1 > evidence/app-pod.log

DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$ kubectl -n "SNS" exec ingestor -- python /opt/s3lab.py verify bench/r1-cl > evidence/before-recovery.jsonl

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$ kubectl -n "SNS" get pod -l app=objects -o yaml > evidence/pod-before.yaml
$ kubectl -n "SNS" get pvc object-data -o yaml > evidence/pvc-before.yaml

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$ kubectl -n "SNS" delete pod "$POD" --wait=false
$ kubectl -n "SNS" rollout status deployment/objects --timeout=120s
deployment "objects" successfully rolled out

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DELL@DESKTOP-UHCRB11 MINGW64 ~/BigData_Lab01_Group4 (main)
$
```

```
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